–28.0

–25

–20

– 15

– 10

–5

0

5

– 10 –5 0 –5

(c)

(a) (b)

x x (mm) (mm)

– 13.0 –12.5
x (mm)

z (mm)

z (mm)

z (mm)

0

–27.5

–27.0

–26.5

–26.0

–25.5

–25.0

–9

– 8

–7

–6

–5

–4

–3

–2
Cardioid

Elecmech

LifeV

OpenCMISS
Simula-FEniCS
Reference configuration

MOOSE-EWE

Hopkins-MESCAL

glasgowHeart-IBFE

CARP

CardioMechanics

Figure 7. Problem2: deformation of a line. Panel (a)shows the deformed location of a line in the middle of the ventricular wall (8.5sin u,0, 18.5cosu), as shown in red in figure2, with details of the apical region (b) and inflection point (c). (Online version
in colour.)

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the deformed positions of the apex at the endo-

on in colour.)

–28.0
–5

(c)

(b)

x (mm)

– 13.0 –12.5
x (mm)

z (mm)

z (mm)

0

–27.5

–27.0

–26.5

–26.0

–25.5

–25.0

–9

–8

–7

–6

–5

–4

–3

–2

–29.0
102 103 104

no. degrees of freedom

epicardial apex

deformed apex locatio
105 106

–28.5

–28.0

–27.5

–27.0

–26.5

BFE
AL

x location. The dashed line separates results for the deformed positions of the apex at the endo-eformed position of the epicardium. (Online version in colour.)

– 10

–5

0

5

(c)

(a) (b)

– 13.0 –12.5
x (mm)

z (mm)

z (mm)

–25.0

–9

–8

–7

–6

–5

–4

–3

–2

tion